Muhammad Dhiaz Rafilianza

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ABOUT ME

Data Scientist with 3+ years of experience, including roles at ByteDance and Tokopedia. Skilled in SQL, BigQuery, Tableau and Python, with experience defining business metrics, building ETL pipelines, conducting A/B tests and mentoring teams. Currently pursuing a Master of Artificial Intelligence at Monash University, with research in video diffusion models and reinforcement learning, and contributing to the Analysis in Medical Imaging team.

EDUCATION

Monash University 2025 – Dec 2026 (Expected)
Master of Artificial Intelligence
Student team: Analysis of Images in Medicine

Sepuluh Nopember Institute of Technology 2016 – 2020
Received Bachelor of Engineering (B.Eng.) in Engineering Physics.
Relevant Courses: Statistics and Stochastic, Computer Programming, Optimization, and Artificial Neural Network.
Thesis: Optimization of Drilling Operation and Control with Stochastic Methods Towards Mud Density Effect with PID Control.

RESEARCH AND ACADEMICS PROJECTS

Improving Geometric Consistency of Video Diffusion Models
Supervisor: Professor Tien-Tsin Wong.
Monash University | Minor Thesis

- Developed joint RGB–semantic video generation using Wan 2.1 and a lightweight 3D CNN.
- Implemented semantic-guided flow correction and RLOO reinforcement learning, including LoRA fine-tuning.
- Built CoTracker and YOLO evaluation pipelines to measure temporal class consistency and per-class performance.

FIT5215 Deep Learning Competition
Monash University

- Achieve top 10 ranking out of 380 with students enrolled in the units.
- Develop and fine tuned a DinoV2 model for classification task, ultimately securing a top-tier leaderboard position.
- Explored and benchmarked multiple vision transformer architectures, including ViT, Swin, BEiT, and DINO to optimize classification performance.

WORK EXPERIENCE

Narasio Data Jakarta, Indonesia
Data Science Mentor July 2024 – February 2025

Narasio Data is a data ecosystem accelerator helping professionals and institutions with their data-driven transformations.

- Promoted data literacy by designing and facilitating comprehensive data science workshops, empowering professionals and institutions to drive data-driven transformations.
- Delivered curriculum covering Inferential Statistics, PySpark, Business Analytics, and Data Visualization, with a specific focus on Data Storytelling to help stakeholders interpret complex metrics.

Bytedance Jakarta, Indonesia
Data Scientist February 2024 – August 2024

ByteDance is a global incubator of platforms at the cutting edge of commerce, content, entertainment and enterprise services - over 2.5bn people interact with ByteDance products including TikTok. Tokopedia was acquired by Bytedance.

- Managed the technical migration of data infrastructure from Tokopedia to ByteDance ecosystems, restructuring Scheduled Queries, Dashboards, and key business metrics.
- Collaborated with User Growth and Operations teams to translate complex business requirements into data-driven solutions and actionable insights.
- Conduct A/B testing on new strategies and refined experimental methodologies to optimize product performance.
- Executed ad-hoc analyses and data validation to track and drive OKR achievement for the user growth division.

Tokopedia Jakarta, Indonesia
Data Analyst September 2021 – August 2024

Tokopedia is one of the leading ecommerce companies in Indonesia, used by millions.

- Spearheaded hyperlocal initiatives across 14 major Indonesian cities, driving sustainability and growth.
- Implemented A/B testing analysis to validate unreleased features, improving the overall user experience.
- Leveraged Google BigQuery for Extract, Transform, and Load (ETL) transforming raw data into actionable insights for informed business decisions.
- **Segmented 1 million+** customers into 6 distinct clusters based on behavior and profitability using K-means clustering, enabling personalized action.
- Developed a product scoring system based on internal and external benchmarks to prioritize product onboarding and optimize warehouse assessments, leading to improved efficiency.
- Develop Machine Learning Platform, enabling **100+ data analytics team** and stakeholders to conduct predictive analysis seamlessly without any coding needed.

Achievement: Data Analyst Mentor Workshop, Data Analyst Social Committee

Rakamin Academy Jakarta, Indonesia
Data Science Mentor November 2021 – February 2023

Rakamin Academy aims to help Indonesians gain access to a more inclusive and impactful education to help them start a career in technology.

- Design and conduct live mentoring sessions for bootcamp students, empowering them with foundational skills in Python, SQL, data processing, data visualization, statistics, and machine learning.
- Provided personalized guidance and rigorous review of homework and final projects for 20+ students, ensuring mastery of key concepts and project requirements.
- Led student teams to exceptional outcomes, with a 50% success rate in securing top or runner-up positions in final project competitions, through strategic guidance, constructive feedback, and unwavering support.

VOLUNTEER AND TEAM EXPERIENCE

Monash AIM
Co-Lead

Monash AIM is an IT student team at Monash University which aims to integrate AI in the medical field to create AI solutions.

- Manage and mentor cross-functional team members, to help them build practical, hands-on skills in PyTorch, Python, and deep learning for medical imaging.
- Co-lead the student team and maintain a strong relationship with the faculty advisor.
- Guide the team's focus and curate project datasets.

AIESEC Indonesia (Surabaya Chapter) Surabaya, Indonesia
Local Committee Vice President, Outgoing Global Talent (International Internships)

AIESEC is the world's largest youth-run organization that aims to develop the leadership potential of young people. Operating in 120+ countries and territories across 2600 universities, we provide students and recent graduates with life-changing opportunities to intern or volunteer abroad within partner organizations.

- Led university outreach and marketing campaigns that resulted in **8 students securing international internships**.
- Managed applicant screening, participant coaching and follow-up after internships.
- Secured a partnership with an international AIESEC chapter to expand internship opportunities for local participants.

GoTo Impact Foundation Jakarta, Indonesia
Data Analyst Mentor June 2022 – November 2023

GoTo Group's non-profit foundation, supporting community development and social-impact initiatives.

- Mentored **100+ students** from technical and non-technical backgrounds in Python, SQL, data visualisation and inferential statistics.
- Delivered live teaching sessions, assessed assignments and provided feedback to strengthen students' practical data analysis skills.
- Guided teams through capstone projects addressing education challenges, helping them analyse data, develop recommendations and present findings to stakeholders.

TECHNICAL SKILLS

Programming: Python, SQL, NumPy, Pandas, Pytorch
Deep Learning and Computer Vision: PyTorch, TensorFlow, vision transformers, image classification, model fine-tuning
Generative AI Research: Diffusions Model
Model Evaluation: Classification benchmarking, adversarial robustness testing
Data Pipelines: Google BigQuery, Airflow, ETL workflows